Present Day, Present Time: AI Exposure & Labor Market Risk

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The Question

November 2022: ChatGPT launches.

Millions of people start using AI to do things that *people get paid to do*.

Some workers are much more exposed than others.

Did it cost them?

Workers differ in how substitutable their tasks are

A Lesson from Trade Economics

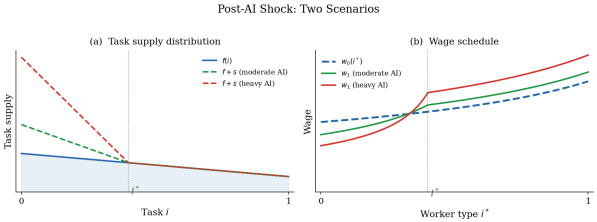
Offshoring (2000s): companies shipped tasks abroad when foreign labor was cheaper

- **Grossman & Rossi-Hansberg (2008):** it's not whole jobs that get offshored — it's specific *tasks*
- **Ottaviano, Peri & Wright (2013):** different workers face different cutoffs — not everyone equally at risk
- **AI is the same story, new technology:**

  Instead of cheap foreign workers — a cheap *machine*.
  "Offshoring" your tasks to Claudeland, GPTopia, Geminia...

The Model: AI as a Supply Shock

Key idea: AI doesn't set prices — it *adds supply* to the tasks it can do



- Every task has human supply $f(i)$. AI adds $s(i)$ on top — but only below I^*
- Two competing forces:
  - ↑ Economy grows ($\bar{Y} > Y_0$) — helps everyone
  - ↓ Extra supply ($f+s > f$) — hurts exposed workers
- **Deep inside** $[0, I^*)$**:** extra supply dominates → wages fall
- **Near or above** I^***:** output growth dominates → wages rise
- As AI improves, I^* shifts right

What the Model Predicts

Workers heavily exposed to AI (s/f large) face two pressures:

Wage compression

AI floods their task market with supply — scarcity falls faster than the economy grows $\rightarrow$ lower pay

Displacement

If the compressed wage drops below what you're willing to accept $\rightarrow$ you exit

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Workers near I^* (mild exposure) can still gain — the output expansion outpaces the small supply addition there. The split is **not** *at I^* but somewhere below it.*

Augmentation channel (AI raising worker productivity) is set aside.

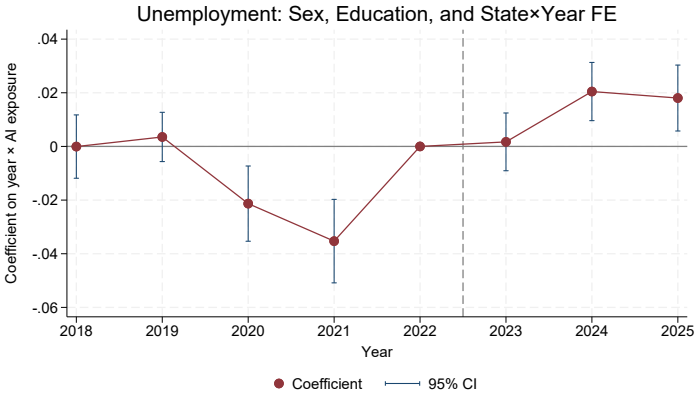
The Strategy

$$Y_{it} = \alpha + \sum_{t \neq 2022} \beta_t \left(\mathbf{1}[t] \times \mathrm{AIExp}_i \right) + X_{it}' \Gamma + \varepsilon_{it}$$

X_{it}: sex, education, state×year FEs
SEs clustered by occupation Baseline: 2022
($\beta_{2022} \equiv 0$)

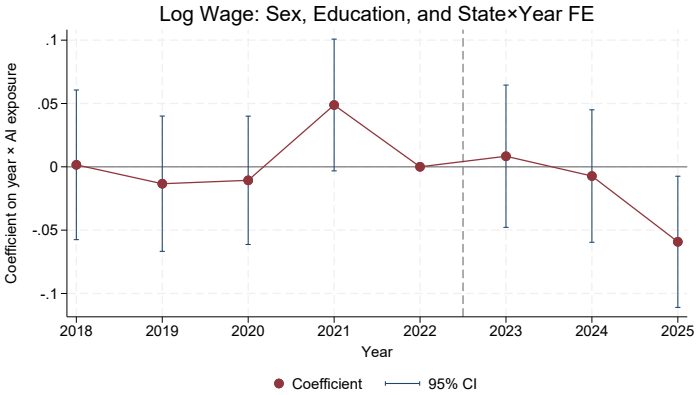
- **Data:** CPS, 500K+ data points, 2018–2025
- **AI exposure:** Eloundou et al. (2024), occupation-level LLM score
- **Strategy:** high vs. low exposure, before vs. after ChatGPT
- **Model says:** β_t diverges post-2022

Result 1: Unemployment



- *Interesting* pre-trends before 2022 — what happened in 2020
- **Post-2022:** high-exposure workers show a rising unemployment gap
- Effect holds across all specifications

Result 2: Wages



- Flat pre-trends — consistent baseline
- **Post-2022:** wages compress for high-exposure workers
- Non-college workers bear the brunt

Thank you!

Questions welcome.

Bottom line:

AI is like offshoring —
some workers face immediate
wage and employment risk

The data says:

Yes, high-exposure workers
are already seeing it —
unemployment *and* potentially wages

Backup: The Dirty Work

Formal model, derivations, and further discussion of I^*

Model Setup (Formal)

- **Tasks** indexed $i \in [0,1]$ by AI susceptibility
  - Low i: routine cognitive — easy for AI — High i: complex — hard for AI *currently*
- **CES Production**:

$$Y = \left[\int_0^1 y(i)^{\frac{\sigma-1}{\sigma}} \, di \right]^{\frac{\sigma}{\sigma-1}}$$

- **Workers:** type i^* specializes in task i^*; supply of task i is $f(i)$ (fixed, inelastic)
- **AI supply** of task i: $s(i) \geq 0, \; s'(i) < 0$ for $i < I^*, \; s(i) = 0$ for $i \geq I^*$
  - AI adds to supply of routine tasks; I^* is the capability frontier
- Competitive firms; output price P_Y fixed (small open economy)

Pre-AI Equilibrium: Derivation

Firm cost minimization gives task demand:

$$y^d(i) \;=\; Y \cdot \left(\frac{P_Y}{p_0(i)}\right)^{\sigma}$$

Market clearing $\left(y^d(i) = f(i)\right)$ pins down task price:

$$p_0(i) \;=\; P_Y \cdot \left(\frac{Y}{f(i)}\right)^{1/\sigma}$$

Solve for Y_0 — plug $y(i) = f(i)$ back into the CES aggregator:

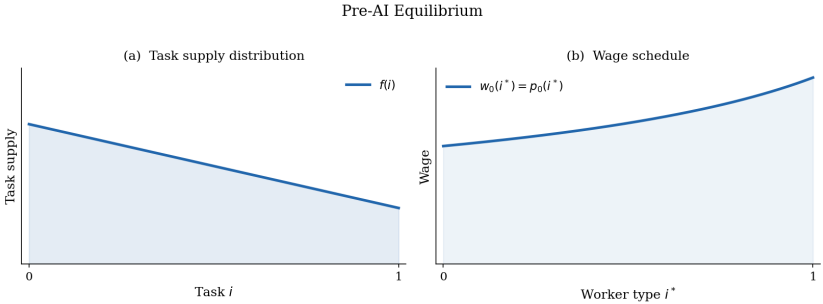
$$\boxed{Y_0 \;=\; \left[\int_0^1 f(i)^{\frac{\sigma-1}{\sigma}}\, di\right]^{\frac{\sigma}{\sigma-1}}}$$

- Y_0 is fully determined by worker distribution f — not by prices

Pre-AI Equilibrium: Wage Schedule

Under full specialization, each worker earns their task price:

$$w_0(i^*) \;=\; p_0(i^*) \;=\; P_Y \cdot \left(\frac{Y_0}{f(i^*)}\right)^{1/\sigma}$$



Supply $f(i)$ is downward-sloping (more workers in routine tasks); wage w_0 mirrors it upward (scarcer tasks pay more).

Post-AI: The Supply Shock

AI adds supply $s(i)$ to tasks below I^*. Effective supply becomes $\bar{y}(i) = f(i) + s(i)$.

New output: $\bar{Y} = \left[\int_0^1 \bar{y}(i)^{(\sigma-1)/\sigma}\, di\right]^{\sigma/(\sigma-1)} > Y_0$

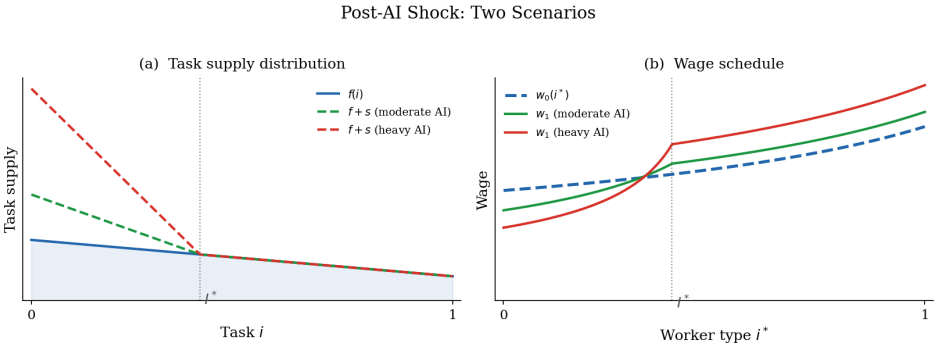
Post-AI wage (same CES logic):

$$\boxed{w_1(i^*) \;=\; P_Y \left(\frac{\bar{Y}}{f(i^*) + s(i^*)}\right)^{1/\sigma}}$$

- **Numerator** $\bar{Y} > Y_0$: economy grew $\rightarrow$ pushes wages up for all
- **Denominator** $f + s > f$ for $i < I^*$: more supply $\rightarrow$ pushes wages down
- For $i \geq I^*$: $s = 0$, only numerator moved $\rightarrow$ wages rise
- For i near I^*: $s(i) \approx 0$, output growth still dominates $\rightarrow$ wages rise
- For i deep inside $[0, I^*)$: s/f large, denominator outpaces numerator $\rightarrow$ wages fall

One channel set aside: *augmentation* (AI raising worker productivity). Left for future work.

Post-AI: Two Scenarios

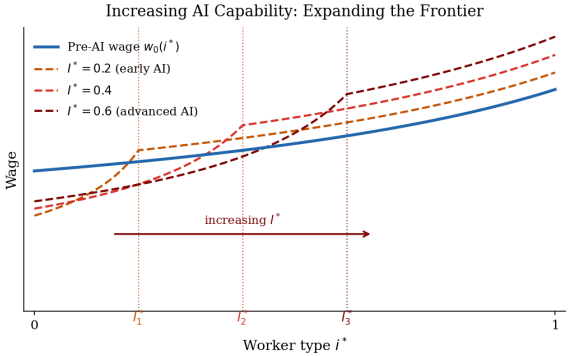


The crossover between gain and compression depends on $s(i)/f(i)$ vs. $\bar{Y}/Y_0 - 1$.
Workers near I^* can gain even below the frontier; deep exposure is needed for compression.

Further Discussion of I^*

As AI improves, I^* shifts right

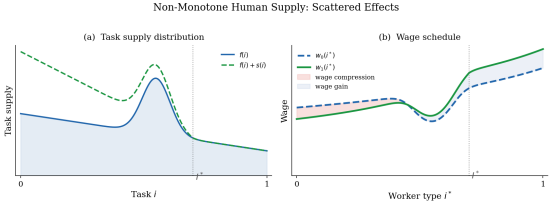
More workers exposed, deeper compression.



Each curve shows w_1 for a different I^*. Higher $I^* \rightarrow$ wider compression.

Non-monotone $f(i)$: scattered effects

If many workers cluster in a mid-complexity task, wage compression need not be contiguous.



Bump in f near $i \approx 0.55$ creates a local dip in w_0 and a pocket where workers may gain even inside $[0, I^*)$.